Design and Development Report

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Render Deployment URL:

https://loan-analysis-application-flask-based.onrender.com

GitHub Repository URL:

https://github.com/showmensarker/Loan-Analysis-Application-Flask-Based

Admin Login Credentials:
Username: admin
Password: admin123

The name of the project is "Loan Analysis Flask Application," and it was built using Flask, a web framework based on the Python language. The application is a web-based tool intended to help manage and analyze data from customers and loans based on a loan default dataset.

In terms of the back-end development, SQLAlchemy serves as ORM, making it easier to handle database transactions. Moreover, SQLite serves as an RDBMS, which facilitates the storage of customer and loan-related data. As for front-end development, HTML, CSS, and Jinja templates have been used.

The software application has all the capabilities for managing customers and their loans. The administrator is capable of viewing, adding, updating, and deleting customers' and loans' information from the software application. Customers' loans history information can be viewed from the software application. Pagination capability was added in the customer and loans section of the software application to facilitate easy interaction with larger amounts of data.

Analysis dashboard was created to generate statistical data. This included the average income for the customer, average loan amount, the highest loan amount issued, total number defaulted loan and the number of high-risk customers. Two charts were designed using Matplotlib for visualizing the loan risk and loan defaults per age group.

The system also incorporates filtering capability, where users can be filtered according to age brackets and minimum credit score. Custom HTTP error pages have been developed for codes 400, 403, 404, 405, and 500.

With regards to security, administrator authentication has been incorporated with the use of Flask's session management tool. Login credentials are needed in order to perform CRUD actions, and there is a feature that allows users to log out of the application.

Testing was done through automation using Pytest in order to make the application more reliable and correct. The tests include route testing, functional testing, authentication testing, pagination testing, validation testing, filtering tests, edge case testing, and HTTP error handling tests. Pytest fixtures were also used to make tests easier to manage. All tests were executed without any problems.

Overall, the Loan Analysis Flask Application demonstrates the development of a secure, modular, and data-centric web application with customer and loan analysis, visualization, authentication, and automated testing features.